

```
vishnu@VISHNU: ~  
GNU nano 8.7.1 shell.c  
#include <stdio.h>  
#include <string.h>  
  
int main() {  
    char input[100];  
  
    while (1) {  
        printf("Vishnu@LAPTOP-SHKTDDI3:~$ ");  
        fgets(input, sizeof(input), stdin);  
  
        input[strcspn(input, "\n")] = '\0';  
  
        if (strcmp(input, "exit") == 0) {  
            printf("Exiting shell...\n");  
            break;  
        }  
  
        printf("You entered: %s\n", input);  
    }  
  
    return 0;  
}  
[ Read 22 lines ]  
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Und  
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_ Go To Line M-E Red
```

1.

Output:

```
vishnu@VISHNU:~$ nano shell.c  
vishnu@VISHNU:~$ gcc shell.c -o shell  
vishnu@VISHNU:~$ ./shell  
Vishnu@LAPTOP-SHKTDDI3:~$ hello  
You entered: hello  
Vishnu@LAPTOP-SHKTDDI3:~$ test  
You entered: test  
Vishnu@LAPTOP-SHKTDDI3:~$ bye  
You entered: bye  
Vishnu@LAPTOP-SHKTDDI3:~$ exit  
Exiting shell...
```

2.

```
vishnu@VISHNU: ~  
GNU nano 8.7.1 keyboard.c  
#include <stdio.h>  
#include <unistd.h>  
#include <termios.h>  
  
int main()  
{  
    struct termios oldt, newt;  
    char ch;  
    tcgetattr(STDIN_FILENO, &oldt);  
    newt = oldt;  
    newt.c_lflag &= ~(ICANON | ECHO);  
    tcsetattr(STDIN_FILENO, TCSANOW, &newt);  
    printf("Type something: ");  
    while(1){  
        ch = getchar();  
        if(ch == '\n'){  
            break;  
        }  
        printf("\nYou pressed: %c", ch);  
        printf("\nType something: ");  
        fflush(stdout);  
    }  
    tcsetattr(STDIN_FILENO, TCSANOW, &oldt);  
    printf("\nDone!\n");  
    return 0;  
}
```

[Read 26 lines]

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Lo
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^_ Go

Output:

```
vishnu@VISHNU:~$ ./keyboard  
Type something:  
You pressed: h  
Type something:  
You pressed: i  
Type something:  
You pressed: e  
Type something:  
You pressed: x  
Type something:  
You pressed: i  
Type something:  
You pressed: t  
Type something: |
```

3.

```
GNU nano 8.7.1 buffer.c
#include <stdio.h>
#include <unistd.h>
#include <termios.h>
#include <string.h>

int main() {
    struct termios oldt, newt;
    char buffer[100];
    int index = 0;
    char ch;

    tcgetattr(STDIN_FILENO, &oldt);
    newt = oldt;
    newt.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSANOW, &newt);

    while (1) {
        index = 0;
        memset(buffer, 0, sizeof(buffer));

        printf("vishnu@VISHNU:~$ ");
        fflush(stdout);

        while (1) {
            ch = getchar();

            if (ch == '\n') {
                buffer[index] = '\0';
                break;
            } else if (ch == 127) {
                if (index > 0) {
                    index--;
                    printf("\b \b");
                    fflush(stdout);
                }
            } else {
                if (index < (int)sizeof(buffer) - 1) {
                    buffer[index++] = ch;
                    printf("%c", ch);
                    fflush(stdout);
                }
            }
        }

        if (strcmp(buffer, "exit") == 0) {

```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify

```

        printf("%c", ch);
        fflush(stdout);
    }
}

if (strcmp(buffer, "exit") == 0) {
    printf("\nExiting shell...\n");
    break;
}

printf("\nYou entered: %s\n", buffer);
}

tcsetattr(STDIN_FILENO, TCSANOW, &oldt);

return 0;
}
|

```

Output:

```

vishnu@VISHNU:~$ nano buffer.c
vishnu@VISHNU:~$ ./buffer
vishnu@VISHNU:~$ hi
You entered: hi
vishnu@VISHNU:~$ helo
You entered: helo
vishnu@VISHNU:~$ exit
Exiting shell...
vishnu@VISHNU:~$ |

```